

CLASS-8

LESSON-3 UNDERSTANDING QUADRILATERALS

(This PDF based on NCERT Book)

CONVEX AND CONCAVE POLYGONS(उत्तल और अवतल बहुभुज)-

A **polygon** is a closed shape with straight sides (like a triangle, square, or pentagon).

Convex Polygons (The "Normal" Ones)

- Think of them as shapes that have **no dents or inward curves**.
- If you rolled a tiny marble around the boundary of a convex polygon, it would always roll **outwards** at every corner.
- **Simple Test:** If you connect any two points *inside* the shape with a straight line, the entire line segment stays **inside** the shape.
- All its interior angles are small (less than 180°).
- **Examples:** A square, a triangle, or a regular stop sign shape.

Concave Polygons (The "Dented" Ones)

- Think of them as shapes that have at least one "**cave**" or "**dent**" pushed inwards.
- If you rolled a tiny marble around the boundary, it would roll **inwards** at the corner with the dent.
- **Simple Test:** You can find at least two points *inside* the shape that, when connected by a straight line, the line segment **passes outside** the shape.
- They must have at least one large interior angle (greater than 180°).
- **Examples:** A crescent moon shape made of straight lines, or a star shape.

In short:

- **Convex:** All corners point **out**.
- **Concave:** At least one corner points **in** (the "cave").

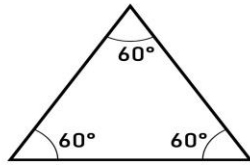
REGULAR AND IRREGULAR POLYGONS(नियमित और अनियमित बहुभुज)-

Regular Polygons (The "Perfect" Shapes)

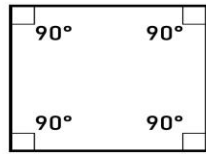
- Think of these as the **symmetrical** shapes where everything is equal.
- **All sides** are the **same length**.
- **All interior angles** are the **same size**.
- A regular polygon is always **convex** (no dents).
- **Simple Test:** If you rotate the shape, it looks exactly the same multiple times before completing a full circle.

- **Examples:** An equilateral triangle (3 equal sides, 3 equal 60° angles), a square (4 equal sides, 4 equal 90° angles), or a regular hexagon.

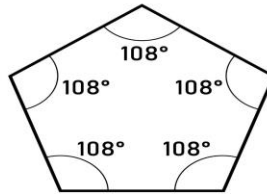
Regular Polygons



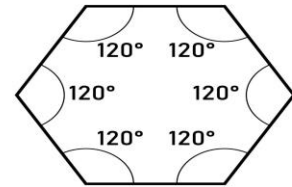
Equilateral Triangle



Square



Regular Pentagon

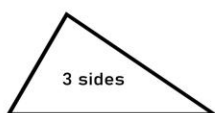


Regular Hexagon

Irregular Polygons (The "Varied" Shapes)

- These shapes are **not uniform**; they have variations.
- **Sides** are **not all the same length**.
- **Angles** are **not all the same size**.
- An irregular polygon can be **convex** (no dents, but uneven sides) or **concave** (has dents).
- **Simple Test:** The shape looks "lopsided" or uneven, and you can easily tell the difference between its sides and angles.
- **Examples:** A rectangle that is not a square, a scalene triangle, or any polygon with a "dent" in it (a concave polygon).

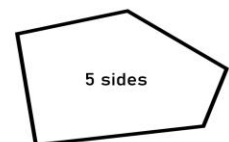
Irregular Polygons



Scalene Triangle



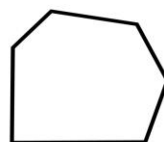
Rectangle



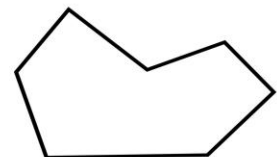
Irregular Pentagon



Irregular Hexagon



Irregular Heptagon



Irregular Octagon

Feature	Regular Polygon	Irregular Polygon
Sides	All are the same length	Not all are the same length
Angles	All are the same size	Not all are the same size
Uniformity	Uniform and symmetrical	Varied or "lopsided"

SUM OF THE MEASURES OF THE EXTERIOR ANGLES OF A POLYGON(किसी

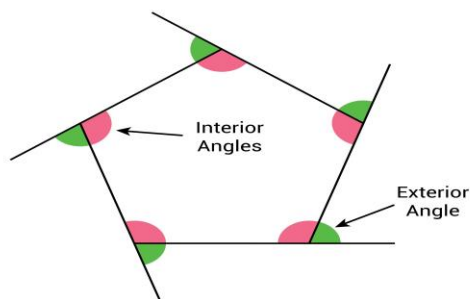
बहुभुज के बाहरी कोणों के मापों का योग)-

The sum of the measures of the exterior angles of **any** convex polygon is always 360° .

Why is it always 360° ?

1. **Imagine Walking Around:** Picture yourself standing at one vertex (corner) of a polygon and walking along its sides. As you turn each corner to walk along the next side, the angle you turn is the **exterior angle**.

Angles of a Polygon



2. **Completing the Circle:** When you finish walking all the way around the polygon and return to your starting point, you have made one complete **full turn**.
3. **Full Turn = 360° :** A complete rotation is always 360° , no matter how many sides the polygon has (whether it's a triangle, a square, or a 100-sided polygon).

Simple Rule:

- **Triangle (3 sides):** Sum of exterior angles = 360°
- **Square (4 sides):** Sum of exterior angles = 360°
- **Octagon (8 sides):** Sum of exterior angles = 360°

This rule holds true for **all convex polygons**.

KINDS OF QUADRILATERALS(चतुर्भुजों के प्रकार)-

A **quadrilateral** is simply any closed shape with **four straight sides**. The kinds of quadrilaterals are grouped based on whether they have parallel sides, equal sides, or right angles (90°).

The Main Groups

The main types of quadrilaterals are:

1. Parallelogram(समांतर चतुर्भुज)

- **What it is:** A shape where **opposite sides are parallel** (run in the same direction and never cross) and **equal in length**.
- **Special Trait:** Opposite angles are also equal.
- **Includes:** Rectangles, Rhombuses, and Squares.

2. Trapezoid (or Trapezium)(समलंब चतुर्भुज)

- **What it is:** A shape that has **only one pair of opposite sides parallel**.
- **Special Trait:** The other two sides can be slanted.
- **Example:** The shape of a water cup where the top is wider than the bottom.

3. Kite

- **What it is:** A shape where **two pairs of adjacent (next to each other) sides are equal** in length.
- **Special Trait:** Looks like a classic diamond-shaped kite that flies in the air. Its diagonals cross at a perfect 90° angle.

ELEMENTS OF PARALLELOGRAM(समांतर चतुर्भुज के तत्व)-

The Parts

1. **Sides (4):** It has four sides.
2. **Corners/Vertices (4):** It has four corners.
3. **Angles (4):** It has four interior angles.
4. **Diagonals (2):** It has two lines connecting opposite corners.

The Rules

The special rules that make it a parallelogram are:

- **Opposite Sides:** The sides across from each other are **parallel** (they never meet) and **equal** in length.
- **Opposite Angles:** The angles across from each other are **equal** in size.
- **Next-Door Angles:** Any two angles that are next to each other always add up to **180°** .
- **Diagonals Rule:** The two lines connecting the corners cross exactly at the **middle** of both lines. They cut each other in half.

The simplest takeaway: If a four-sided shape has **opposite sides that are both parallel and equal**, it's a parallelogram.

ANGLES OF PARALLELOGRAM(समांतर चतुर्भुज के कोण)-

1. Opposite Angles are Equal

The angles that are **directly across** from each other are always the **same size**.

- *Example:* If the top-left angle is 110° , the bottom-right angle is also 110° .

2. Next-Door (Consecutive) Angles Add up to 180°

Any two angles that are **side-by-side** (next to each other) must always add up to 180° .

- *Example:* If one angle is 110° , the angle next to it must be $180^\circ - 110^\circ = 70^\circ$.

3. All Four Angles Add up to 360°

Like any four-sided shape (quadrilateral), the sum of all four interior angles is always 360° .

DIAGONALS OF PARALLELOGRAM(समांतर चतुर्भुज के विकर्ण)-

The diagonals of a parallelogram are the **two straight lines** you draw inside the shape that connect its **opposite corners**.

These diagonals have one main, important rule:

- They **bisect** each other, which means they **cut each other exactly in half**.

Rule Means:

Imagine the two diagonals crossing each other inside the parallelogram.

1. The point where they cross is the **exact middle** of the entire parallelogram.
2. The point of intersection cuts the first diagonal into two **equal pieces**.
3. The point of intersection also cuts the second diagonal into two **equal pieces**.

In short: They meet right in the middle, and they *always* cut each other in half.

SOME MORE INTERESTING KINDS OF PARALLELOGRAM(समांतर चतुर्भुज के कुछ और रोचक प्रकार)-

These shapes are special because they follow the rules of a parallelogram *plus* an extra rule:

Quadrilateral	Simple Definition	Key Properties
Rectangle(आयत)	A parallelogram with four right angles (all 90°).	Opposite sides are equal and parallel.
Rhombus(विषमकोण)	A parallelogram with all four sides equal in length.	The diagonals cross at a 90° angle.
Square(वर्ग)	A shape that is both a Rectangle AND a Rhombus .	All four sides are equal, and all four angles are 90° .